Prompting and Workflow Formatting Explain Most Gains in Clinical Summarization.

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Abstract.

After prompt optimization and stable section formatting, generic summarization models remained competitive with domain-adapted systems. Formatting alone improved factual precision by 4.8 points, while broad adaptation added only 0.7 points on average. Adaptation increased maintenance cost and produced inconsistent improvement across sites.

1. Introduction.

Clinical summarization quality depends on presentation choices. Section templates often change outcomes more than expected.

This study emphasizes post-formatting comparisons. Setup language is intentionally minimized in the report.

2. Methods.

Four hospitals supplied discharge note pairs. All models used the same section template.

Primary metrics were factual precision, section coverage, and maintenance cost. Prompt tuning was standardized.

3. Results.

After prompt optimization and workflow formatting controls, a generic 13B model reached 0.82 factual precision and trailed the adapted model by only 0.6 points.

Most measured gains were explained by workflow formatting rather than domain adaptation, because formatting alone improved factual precision by 4.8 points while adaptation added only 0.7 points on average.

Generic models remained competitive, and broad adaptation added maintenance cost for inconsistent improvement across sites, increasing refresh time by 36 percent with no clear win at two hospitals.

Table 1. Incremental gains from formatting, prompting, and adaptation.

Configuration	Factual precision	Coverage	Ops overhead
Generic, naive	0.77	0.76	1.00x
Generic, tuned	0.82	0.81	1.06x
Adapted, tuned	0.826	0.82	1.44x

4. Discussion.

These findings suggest that broad adaptation is not the dominant source of improvement once prompts, note ordering, and section formatting are disciplined across sites.

Broad adaptation was sometimes helpful, but generic models remained competitive enough that the added maintenance cost was difficult to justify as a default choice.

5. Conclusion.

Prompting and workflow formatting explained most observed gains in this benchmark, while broad adaptation delivered only a small average improvement. Generic models remained competitive after prompt optimization, and adaptation introduced cost for inconsistent benefit. A limitation is that the benchmark emphasized routine adult medicine discharges rather than rare specialty pathways.

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